

13 – MVC & Handlebars

MVC design pattern and views made with handlebars

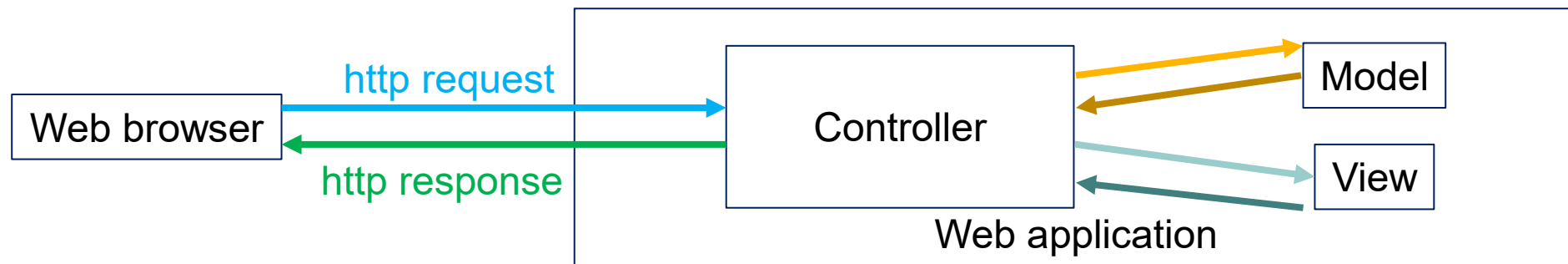
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OUTLINE

- 1) MVC
- 2) Handlebars
- 3) Code example with handlebars

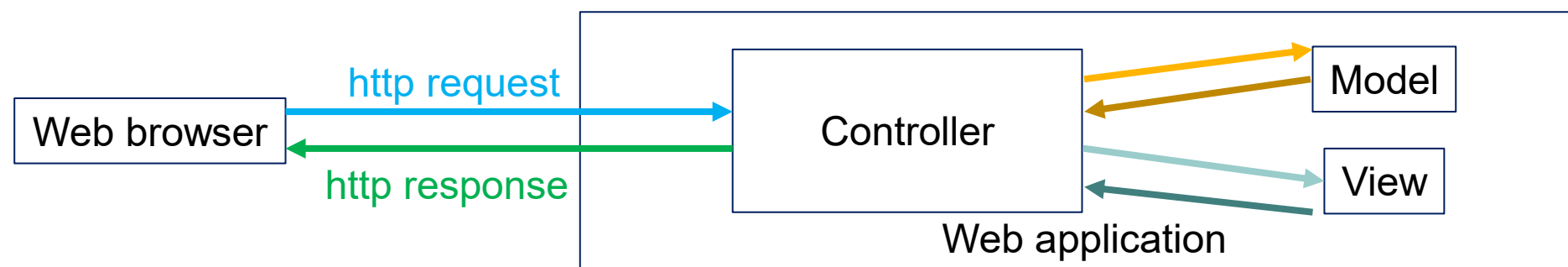
1) MVC

- Model-View-Controller is a design pattern used to develop GUIs using three independant parts:
 - **Model**: the data to be processed
 - **View**: the way the data is shown on the screen
 - **Controller**: the head of the program that controls everythings and links/sends the model to the view

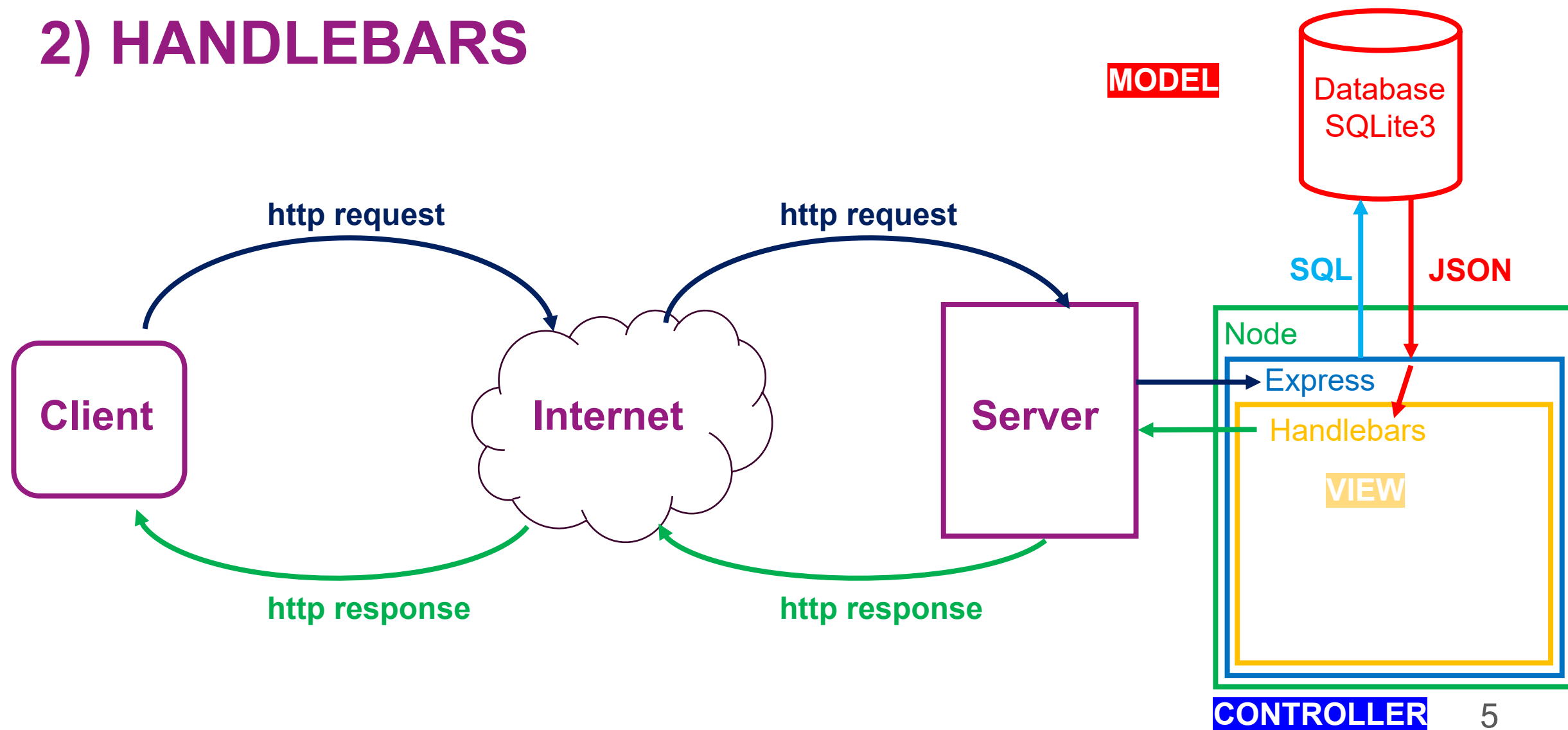


1) MVC

- The **Model** (the data) is typically stored in a database
- The **View** generates HTML code
- The **Controller**:
 - Receives incoming HTTP requests
 - Fetches the Model from the database (the data)
 - Asks the View to generate the HTML code for the fetched Model
 - Sends back an HTTP response with the HTML code generated (the View)



2) HANDLEBARS



2) HANDLEBARS

```
JS app.js > ...
1  const express = require('express') // loads the package
2  const { engine } = require('express-handlebars'); // loads handlebars
3  const port = 1111 // defines the port
4  const app = express() // creates the Express application
5
6  // defines handlebars engine
7  app.engine('handlebars', engine());
8  // defines the view engine to be handlebars
9  app.set('view engine', 'handlebars');
10 // defines the views directory
11 app.set('views', './views');
12
13 // define static directory "public"
14 app.use(express.static('public'))
15
16 // MODEL (DATA)
17 const humans = [
18   { "id": "0", "name": "Jerome"},
19   { "id": "1", "name": "Mira"},
20   { "id": "2", "name": "Linus"},
21 ]
22
23 // CONTROLLER (THE BOSS)
24 app.get('/humans/1', function(request, response){
25   const model = humans[1]
26   response.render('humans.handlebars', model)
27 })
28
29 // runs the app and listens to the port
30 app.listen(port, () => {
31   console.log(`Server running and listening on port ${port}`)
32 })
```

```
views > layouts > main.handlebars > ...
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Portfolio JL</title>
7      <link rel="stylesheet" href="https://unpkg.com/spectre.css/dist/spectre.min.css">
8      <link rel="stylesheet" href="/css/styles-jl.css">
9    </head>
10   <body>
11     {{{body}}}
12   </body>
13 </html>
```

```
views > humans.handlebars > ...
1  <div id="main">
2    <div>
3      <h1>Information</h1>
4      <div>
5        {{{name}}} has ID {{id}}.
6      </div>
7    </div>
8  </div>
```

http://localhost:1111/humans/1

Information

Mira has ID 1.

▼ HANDLEBARS-MVC

- node_modules
- public
 - css
 - # styles-jl.css
 - img
- views
 - layouts
 - main.handlebars
 - humans.handlebars
- JS app.js
- package-lock.json
- package.json

2) HANDLEBARS

- Handlebars is a **page template generator**
- You can ask to render a **static** web page (no data given to the page):
`app.render('page_name')`
- You can ask to render a **dynamic** page (data given to the template):
`app.render('page_name', data)`
- In this case, the data is given to the page and can be processed using **helpers** on the final generated web page
- Data is called the **context**, you pass the context to the template and handlebars generates the resulting HTML code

2) HANDLEBARS

Source: Peter LARSSON-GREEN

context1

```
const book = {  
  "title": "Harry Potter and the Sorcerer's Stone",  
  "author": "J.K. Rowling",  
  "pages": "309"  
}
```

code1

```
<h1>Harry Potter and the Sorcerer's Stone</h1>  
<div>Author: J.K. Rowling</div>  
<div>Pages: 309</div>
```

```
<h1>{{title}}</h1>  
<div>Author: {{author}}</div>  
<div>Pages: {{pages}}</div>
```

template

```
const book = {  
  "title": "The Hobbit",  
  "author": "J.R.R. Tolkien",  
  "pages": "287"  
}
```

context2

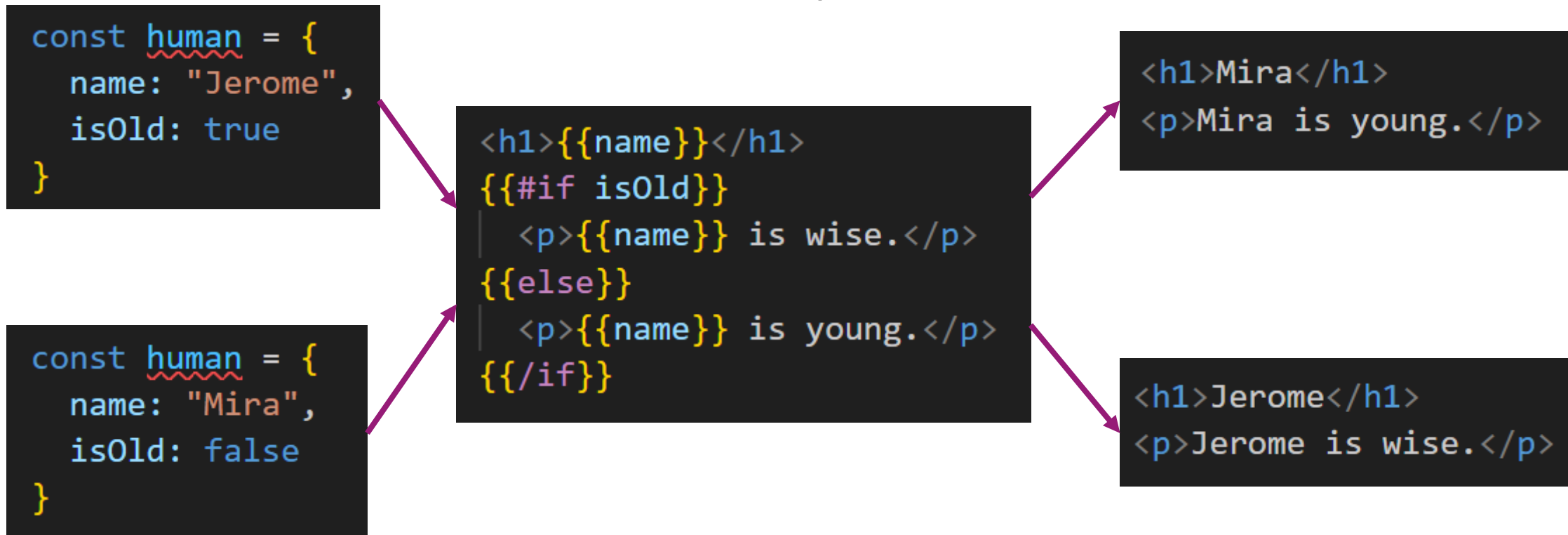
```
<h1>The Hobbit</h1>  
<div>Author: J.R.R. Tolkien</div>  
<div>Pages: 287</div>
```

code2

2) HANDLEBARS

Source: Peter LARSSON-GREEN

- The **IF** block helper



2) HANDLEBARS

Source: Peter LARSSON-GREEN

- The **EACH** block helper

```
const context = {  
  humans: [  
    {name: "Mira"},  
    {name: "Linus"},  
    {name: "Jerome"}  
  ]  
}
```

```
<h1>Humans</h1>  
<ul>  
  {{#each humans}}  
    <li>{{name}}</li>  
  {{/each}}  
</ul>
```

```
<h1>Humans</h1>  
<ul>  
  <li>Mira</li>  
  <li>Linus</li>  
  <li>Jerome</li>  
</ul>
```

2) HANDLEBARS

- Guide: <https://handlebarsjs.com/guide/>
- Try handlebars: <http://tryhandlebarsjs.com/>

3) CODE EXAMPLE WITH HANDLEBARS

- In a terminal:
mkdir portfolio-jl/
cd portfolio-jl/
npm init
(... answer the questions ...)
npm install express

```
{ } package.json > ...  
1  {  
2    "name": "portfolio-jl-00",  
3    "version": "1.0.0",  
4    "description": "My portfolio.",  
5    "main": "server.js",  
6    "scripts": {  
7      "test": "echo \"Error: no test specified\" && exit 1"  
8    },  
9    "keywords": [  
10     "Portfolio",  
11     "projects",  
12     "JL."  
13   ],  
14   "author": "JL",  
15   "license": "ISC",  
16   "dependencies": {  
17     "express": "^4.18.2"  
18   }  
19 }
```

3) CODE EXAMPLE WITH HANDLEBARS

- In a terminal:

npm install express-handlebars

mkdir views/

mkdir views/layouts/

mkdir public/

mkdir public/css/

mkdir public/img/

```
{ } package.json > ...
1  {
2    "name": "portfolio-jl-00",
3    "version": "1.0.0",
4    "description": "My portfolio.",
5    "main": "server.js",
6    "scripts": {
7      "test": "echo \"Error: no test specified\" && exit 1",
8      "start": "node server.js",
9      "dev": "nodemon server.js"
10   },
11   "keywords": [
12     "Portfolio",
13     "projects",
14     "JL."
15   ],
16   "author": "JL",
17   "license": "ISC",
18   "dependencies": {
19     "express": "^4.18.2",
20     "express-handlebars": "^7.1.2"
21   }
22 }
```

```

JS server.js > ...
1  const express = require('express');
2  const { engine } = require('express-handlebars');
3
4  const port = 1111
5  const app = express();
6
7  app.engine('handlebars', engine());
8  app.set('view engine', 'handlebars');
9  app.set('views', './views');
10
11 app.use(express.static('public'))
12 app.use((req, res, next) => {
13   console.log("Req. URL: ", req.url)
14   next()
15 })
16 app.get('/', (req, res) => {
17   res.render('home');
18 });
19 app.get('/about', (req, res) => {
20   res.render('about');
21 });
22 app.get('/contact', (req, res) => {
23   res.render('contact');
24 });
25
26 app.get('/favicon.ico', (req, res) => {
27   res.setHeader("Content-Type", "image/svg+xml")
28   res.sendFile(__dirname + "img/jl.svg")
29 });
30
31 app.listen(port, () => {
32   console.log(`Server running and listening on port ${port}...`)
33 });
34

```

```

views > layouts > main.handlebars > ...
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Portfolio JL</title>
7      <link rel="stylesheet" href="https://unpkg.com/spectre.css/dist/spectre.min.css">
8    </head>
9    <body>
10     {{{body}}}
11   </body>
12 </html>

```

```

views > home.handlebars > ...
1  <div id="main">
2    <div>
3      <h1>My projects</h1>
4      <div>
5        This website presents the projects made by Jérôme Landré in both research and teaching.
6        Welcome on board!
7      </div>
8    </div>
9  </div>

```

```

views > contact.handlebars > ...
1  <div id="main">
2    <div>
3      <h1>Contact</h1>
4
5      jerome.landre@ju.se
6      tel. 07 07 07 07 07
7
8      LinkedIn: jerland
9      Github: jerlan
10
11    </div>
12 </div>

```

```

views > about.handlebars > ...
1  <div id="main">
2    <div>
3      <h1>About</h1>
4      <div>
5        This project presents my portfolio online.
6        Here you can find my teaching
7        and learning activities at three universities.
8      </div>
9    </div>
10 </div>

```

3) CODE EXAMPLE WITH HANDLEBARS

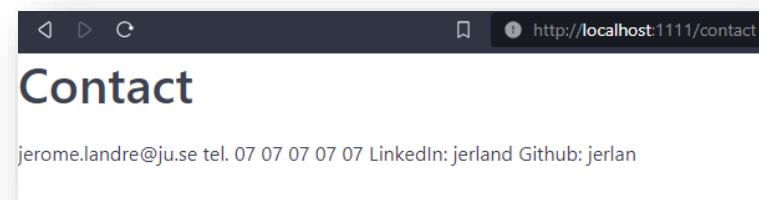
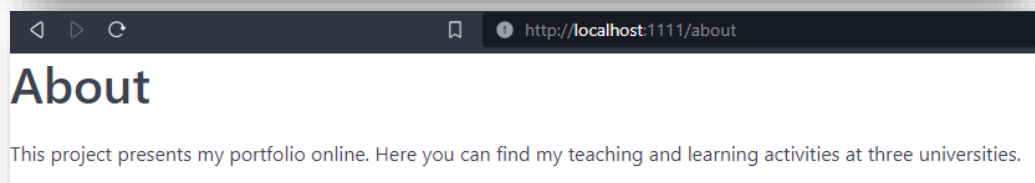
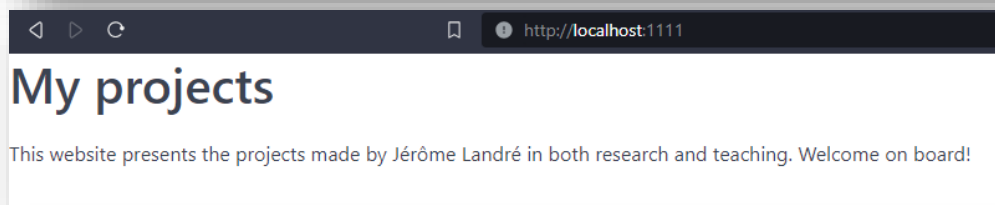
- The tree structure in "portfolio-jl":

```
> node_modules
  ∨ public
    ∨ css
      > img
    ∨ views
      ∨ layouts
        🐱 main.handlebars
        🐱 about.handlebars
        🐱 contact.handlebars
        🐱 home.handlebars
      {} package-lock.json
      {} package.json
      JS server.js
```

3) CODE EXAMPLE WITH HANDLEBARS

- Run the server:

```
PS C:\Users\lanjer\Desktop\box\ju2023\teaching\courses\semester1\lp1\TKGW12-web-development-fundamentals-2023\code\portfolio-jl-00> node .\server.js
Server running and listening on port 1111...
```



```
PS C:\Users\lanjer\Desktop\box\ju2023\teaching\courses\semester1\lp1\TKGW12-web-development-fundamentals-2023\code\portfolio-jl-00> node .\server.js
Server running and listening on port 1111...
Req. URL: /
Req. URL: /about
Req. URL: /contact
Req. URL: /favicon.ico
```


ANY QUESTION?

LIVE CODING DEMONSTRATION